Hithesh Rai Purushothama

480-289-0708 | hraipuru@asu.edu | linkedin.com/in/hithesh-rai-p | github.com/hitheshrai | hitheshrai.github.io/Hithesh

Professional Summary

Materials and AI engineering graduate student with hands-on research experience spanning photovoltaic materials, electrochemical impedance spectroscopy, and physics-informed machine learning across four international research institutions. Builds end-to-end systems across the full stack: from thin-film device fabrication and EIS characterization to GPR-based degradation prediction pipelines, edge AI deployment on constrained hardware, and autonomous robotic systems. Equally comfortable in a materials science lab, an HPC environment, or a robotics integration workflow. Interested in self-driving laboratory systems, transferable ML frameworks for energy materials, and deploying AI in low-connectivity or resource-constrained environments.

Education

Arizona State University — Tempe, AZ
M.S. in Artificial Intelligence Engineering (Materials Science and Engineering) — Jan 2026 – May 2027

Arizona State University — Tempe, AZ
B.S.E. Electrical and Electronic Engineering — Aug 2021 – Dec 2025

Research Experience

Renewable Energy Materials and Devices Lab, Arizona State University — Tempe, AZ
Research Assistant — Sep 2022 – Present

- Developing an ML pipeline using Gaussian Process Regression (GPR) on electrochemical impedance spectroscopy (EIS) features to predict battery capacity fade and interfacial degradation, with application to state-of-health estimation across energy storage systems.
- Reproduced and extended Zhang et al. (2020, *Nature Communications*) battery degradation model, achieving $R^2 = 0.81$ on unseen test cells; identified a 17.80 Hz EIS feature as a degradation-robust predictor across temperature regimes using ARD kernel weighting.
- Applying multitask and transfer learning to model interfacial degradation across batteries, perovskite solar cells, and fuel-cell-relevant material systems from shared EIS descriptors.
- Automated EIS data preprocessing, Distribution of Relaxation Times (DRT) peak extraction, and GPR model training pipeline in Python (NumPy, GPyTorch, scikit-learn) on ASU's Sol HPC cluster via VSCode remote integration.
- Fabricated and optimized perovskite thin films using spin coating and blade coating on rigid and flexible substrates, achieving uniform films with reduced defect density.
- Conducted optical and electrical characterization with solar simulator, XRD, and profilometry to analyze degradation and identify key failure mechanisms guiding material improvements.
- Presented work at IEEE PVSC 2024 (Seattle) and IPERO 2025 (Kyoto); submitted abstract to AI4X – Accelerate Conference 2026 (Singapore) with advisor N. Rolston.

Photovoltaics Lab, École Polytechnique Fédérale de Lausanne (EPFL) — Neuchâtel, Switzerland
Undergraduate Research Assistant — ThinkSwiss Scholar — May 2025 – Aug 2025

- Fabricated single-junction perovskite thin-film devices achieving efficiencies up to 19% using controlled spin-coating workflows with systematic process parameter variation.
- Deposited SnO_2 electron transport layers via ALD and analyzed thickness and conformity; used thermal evaporation for transport layers and metal contacts, improving device yield.
- Assisted with encapsulation and stability testing, identifying key degradation pathways and informing process-improvement actions.

Helmholtz Zentrum Berlin — Quantum Materials Group — Berlin, Germany
International Summer Student — Jul 2024 – Aug 2024

- Investigated structural instabilities in perovskite-based ferroelectric and piezoelectric systems using Pair Distri-

bution Function (PDF) analysis of X-ray and neutron diffraction data.
- Identified composition-driven morphotropic phase transitions and correlated atomic-scale structural variations with compositional changes using Python analysis scripts.

Letian Dou Group, Purdue University West Lafayette, IN
Summer Undergraduate Research Fellow May 2023 – Aug 2023

- Supported halide perovskite stability studies through device fabrication and testing; reviewed and organized five years of literature on additive engineering and degradation mechanisms.
- Compiled a comparative Python database of device structures, processing conditions, and efficiencies; used the Perovskite Database ML platform to identify material-performance trends guiding additive selection.

Engineering Experience

Capstone Project — Power System Modeling for Grid-Scale Energy Storage Tempe, AZ
Arizona State University Jan 2025 – Present

- Modeled Battery Energy Storage Systems (BESS) for renewable energy integration using PyPSA, Pandas, and NumPy, producing performance summaries that informed storage sizing and operational trade-off decisions.
- Deployed an LLM-powered digital twin dashboard to monitor and interpret real-time PV + BESS telemetry, enabling early fault detection and interpretable system diagnostics.
- Designed realistic grid-scale load profiles from utility and consumer datasets to evaluate system-level PV and storage performance.

Leadership & Management

Next Lab, Arizona State University Tempe, AZ
Management Intern Mar 2026 – Present

- Lead evaluation and selection of emerging AI technologies for lab adoption, assessing technical maturity, integration feasibility, and deployment cost-benefit across candidate systems — serving as the lab's primary technology intelligence function.
- Drive end-to-end AI project execution from scoping through implementation, coordinating across research and operations staff to deliver production-ready systems on schedule.
- Support core lab leadership on strategic initiatives, translating complex technical findings into actionable recommendations for non-technical stakeholders.

Next Lab, Arizona State University Tempe, AZ
Senior Studio Associate — AI & Edge Systems Sep 2023 – Present

- Built end-to-end benchmarking pipelines in Python to evaluate AI inference systems across latency, memory, accuracy, hallucination rate, and power consumption on NVIDIA Jetson devices, informing deployment decisions for production edge environments.
- Analyzed trade-offs between INT8/Q4 quantization strategies and model performance, identifying failure modes and guiding optimization for robust real-world deployment at scale.
- Designed and validated agent-based task execution and knowledge-retrieval pipelines using LangChain, reducing manual workflow overhead and improving automation reliability.
- Deployed a digital twin dashboard powered by an LLM to monitor and interpret PV + BESS system telemetry in real time, enabling early fault detection across operational scenarios.

Engineering Projects in Community Service (EPICS), Arizona State University Tempe, AZ
Team Lead Jan 2022 – Jan 2024

- Led a team of 10+ students to develop a Python and robotics curriculum for middle schoolers across the US and South Africa, delivering hands-on embedded AI modules using NVIDIA Jetson hardware with Jetson Infinity.
- Applied NumPy and Matplotlib for simulation components; enhanced learning outcomes through robotic arm hands-on activities.

Selected Projects

NEXUS — Automated Research Intelligence System Next Lab, ASU

- Built an automated morning briefing pipeline delivering daily AI/energy research summaries to WhatsApp using local LLMs on an NVIDIA DGX Spark, with a rolling context file system and automated references pipeline for citation-grounded output.
- Deployed and debugged NVIDIA NIM containers on DGX Spark, resolving permission, Docker socket, and NGC authentication issues; implemented Tailscale for secure remote access.

Nex Bot — Autonomous Voice Receptionist Next Lab, ASU

- Deployed Reachy Mini humanoid robot as an AI-powered voice receptionist with full inference offloaded to a DGX Spark over Tailscale, enabling low-latency on-device interaction without cloud dependency.
- Integrated faster-whisper (STT), Silero VAD, Kokoro TTS, and a local LLM into a real-time conversational pipeline using the `reachy-mini` PyPI stack.

Autonomous Social Robot System Next Lab, ASU

- Built a fully offline public-facing autonomous robot using Reachy Mini + DGX Spark, integrating YOLOv8n for person detection, Silero VAD for voice activity detection, and a complete STT → LLM → TTS pipeline running without internet connectivity.
- Resolved Zenoh daemon connectivity and Docker stack configuration issues to achieve stable real-time inference across all modalities.

EIS-GPR Battery Degradation Pipeline Rolston Lab, ASU

- Reproduced Zhang et al. (2020) in both scikit-learn and GPyTorch; extended to multi-temperature training with ARD kernels, isolating SEI-layer feature at 17.80 Hz as a cross-temperature degradation predictor ($R^2 = 0.81$ on held-out test cells).
- Documented full methodology in a CLAUDE.md session file; planning Fig. 1–2 reproduction using 25°C test cells and extension to DRT-derived features.

Conferences and Presentations

- AI4X – Accelerate Conference 2026, Singapore (accepted) — "Transferable Impedance-Grounded Learning for Interfacial Degradation Across Energy Systems" (with N. Rolston).
- IEEE Photovoltaic Specialists Conference, June 2024, Seattle, WA — "Open-Air Processing of Cesium-Based Inorganic Perovskites for Radioisotopes."
- International Conference on Perovskite, Organic Photovoltaics and Optoelectronics, Jan 2025, Kyoto, Japan — "Blade-Coated Cesium Lead Halide Perovskite Thin Films for Alphavoltaic Applications."
- Stanford Research Conference, April 2024, San Jose, CA — "Enhancing the Performance of Formamidinium Lead Halide Perovskite Solar Cells."
- Arizona Student Energy Conference (AzSEC), April 2023, Tempe, AZ — "Unlocking the Potential of Halide Perovskites: The Art of Thin Film Fabrication."

Awards

- **ThinkSwiss Research Scholarship**, Swiss National Science Foundation Feb 2025
  Awarded for summer research at EPFL on perovskite thin-film device fabrication and stability studies. Washington, DC
- **Fulton Undergraduate Research Initiative Grant (FURI)** Fall/Spring 2023–2025
  Arizona State University, Tempe, AZ
- **Purdue Summer Undergraduate Research Fellowship** Summer 2023
  Purdue University, West Lafayette, IN

Technical Skills

- **ML & Data:** Gaussian Process Regression, Transfer/Multitask Learning, Time-Series Forecasting, Bayesian Modeling, scikit-learn, GPyTorch, TensorFlow, NumPy, Pandas
- **Energy Systems:** EIS, DRT Analysis, PAIOS, PyPSA, PVlib, GridLAB-D, MATLAB/Simulink, BESS Modeling, Load Flow Analysis, SCADA/EMS Basics
- **Fabrication:** Spin Coating, Blade Coating, ALD, Thermal Evaporation, Encapsulation

- **Characterization:** Solar Simulator, XRD, SEM, Profilometry, Optical Inspection, Pair Distribution Function (PDF) Analysis
- **Edge AI & Inference:** ONNX Runtime, TensorRT, MLC, INT8/Q4 Quantization, NVIDIA Jetson, NVIDIA DGX Spark, YOLOv8n
- **Robotics & Autonomy:** Reachy Mini (`reachy-mini`), faster-whisper, Silero VAD, Kokoro TTS, Zenoh, LangChain
- **Infrastructure:** Docker, GitHub, Tailscale, Embedded Linux (Jetson/Pi/DGX), HPC (Sol/SLURM), VSCode Remote
- **Programming:** Python, C++, MATLAB, Verilog
- **EDA Tools:** ModelSim, Quartus